Constructions of Self-Excited and Hidden Attractors with Prescribed Lyapunov Dimension Properties

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Abstract

We give two constructions based on the suspension of a hyperbolic toral automorphism. The first is a self-excited attractor of Lyapunov dimension 10/3 whose only unstable equilibrium has dimension 3/2. All finite-time singular-value exponents converge uniformly on the attractor, so the adjacent cumulative-exponent maxima have a common critical point. The second is a hidden attractor of dimension 7/2 with dense unstable periodic orbits, none of which attains its maximal local Lyapunov dimension. Both attracting sets support transitive hyperbolic flows.

1 Preliminaries and statements

Let X be a smooth vector field on an open set $U \subseteq \mathbb{R}^n$, with forward-complete flow φ^t. For a compact invariant set A, write

$$\mathcal{B}(A) = \{u \in U : \operatorname{dist}(\varphi^t(u), A) \longrightarrow 0\}.$$

An attractor will mean a nonempty compact invariant set having a compact positively invariant neighborhood which it attracts uniformly. The attractors constructed below are also transitive and inclusion-minimal among compact local attractors.

Definition 1. An attractor A is self-excited if some equilibrium belongs to $\overline{\mathcal{B}(A)}$. It is hidden if every equilibrium has a neighborhood disjoint from $\mathcal{B}(A)$. A nonstationary periodic orbit is unstable if it is not Lyapunov stable as an invariant set.

Let $\sigma_1(t,u) \geq \cdots \geq \sigma_n(t,u) > 0$ be the singular values of the full ambient derivative $D\varphi^t(u)$, and put

$$\nu_i(t,u) = t^{-1} \log \sigma_i(t,u), \qquad S_j(t,u) = \sum_{i=1}^{j} \nu_i(t,u), \quad S_0(t,u) = 0.$$

Set $j(t,u) = \max\{0 \leq j \leq n : S_j(t,u) \geq 0\}$. The finite-time local dimension is

$$\mathrm{LD}(t,u) = \begin{cases} j(t,u) + S_{j(t,u)}(t,u)/|\nu_{j(t,u)+1}(t,u)|, & j(t,u) < n, \\ n, & j(t,u) = n. \end{cases}$$

In particular it is zero when the first exponent is negative. We distinguish the local and set quantities

$$\mathrm{LD}(u) = \limsup_{t\to\infty} \mathrm{LD}(t,u), \qquad \dim_{\mathrm{L}} A = \liminf_{t\to\infty} \sup_{u\in A} \mathrm{LD}(t,u). \tag{1}$$

For $d = j + s$, $0 \leq s < 1$, define $\omega_d(L) = \sigma_1(L)\cdots\sigma_j(L)\sigma_{j+1}(L)^s$, with $\omega_0 = 1$ and $\omega_n = |\det L|$. We also consider the exponential-growth definition

$$d_{\mathrm{L}}^{E}(u) = \inf\left\{ d \in [0,n] : \limsup_{t\to\infty} t^{-1} \log \omega_d(D\varphi^t(u)) < 0 \right\}, \tag{2}$$

assigning value n if the set is empty. These definitions need not agree in general. The calculations below give the required comparisons directly.

Definition 2. The common-critical-point condition on A, at the dimension crossing index d_0, is the existence of $u_* \in A$ such that

$$\lim_{t\to\infty} S_j(t,u_*) = \lim_{t\to\infty} \max_{u\in A} S_j(t,u), \qquad j = d_0, d_0+1. \tag{3}$$

Both limits are required to exist. This is the condition in [1, Corollary 6.8, equation (6.58)].

The self-excited assertion considered here is that, for an attractor satisfying (3), $\dim_{\mathrm{L}} A \le \mathrm{LD}(p)$ for some unstable equilibrium p. The hidden assertion is that $\sup_{u\in A} \mathrm{LD}(u)$ is attained on an unstable periodic orbit in A. The following theorems contradict these respective assertions.

Theorem 3. *There is a smooth forward-complete vector field on an open $U \subset \mathbb{R}^6$ with exactly two equilibria and a compact transitive hyperbolic self-excited attractor A such that*

$$\dim_{\mathrm{L}} A = \mathrm{LD}(u) = d_{\mathrm{L}}^E(u) = \frac{10}{3} \quad (u \in A).$$

Its only unstable equilibrium p satisfies $\mathrm{LD}(p) = d_{\mathrm{L}}^E(p) = 3/2$. A branch of $W^u(p)$ has omega-limit set A. Condition (3) holds at every point of A with $d_0 = 3$.

Theorem 4. *For some finite n, there is a smooth complete vector field on $\mathbb{R}^n$ with a compact global attractor and a compact transitive hyperbolic hidden local attractor A such that*

$$\dim_{\mathrm{L}} A = \max_{u\in A} \mathrm{LD}(u) = \max_{u\in A} d_{\mathrm{L}}^E(u) = \frac{7}{2}.$$

Unstable periodic orbits are dense in A, but every periodic point $u \in A$ satisfies $\mathrm{LD}(u) = d_{\mathrm{L}}^E(u) < 7/2$.

In both theorems A is a smooth three-manifold. No noninteger Hausdorff-dimension hypothesis is imposed. In Theorem 3, condition (3) is asserted on A, not on every invariant set or on a different global attractor. Thus the terminology "typical" is used only through the explicit condition above, not through a Baire-generic interpretation.

2 The suspension

Put

$$B = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \qquad \Lambda = \frac{3+\sqrt{5}}{2}, \qquad \lambda = \log\Lambda.$$

Let $f(x) = Bx$ on $\mathbb{T}^2$ and let

$$M = (\mathbb{T}^2 \times \mathbb{R})/((x, s+1) \sim (f(x), s)), \qquad \psi^t[x,s] = [x, s+t].$$

Choose unit eigenvectors v_u, v_s of B, corresponding to Λ, Λ^{-1}.

Lemma 5. *The suspension is a nonsingular transitive Anosov flow with dense unstable periodic orbits. There is a smooth metric g on M for which the singular values of $D\psi^t$ are exactly $e^{\lambda t}, 1, e^{-\lambda t}$ at every point.*

Proof. On the covering space put

$$E_u = e^{-\lambda s} v_u, \qquad E_0 = \partial_s, \qquad E_s = e^{\lambda s} v_s.$$

The deck transformation $(x, s) \mapsto (Bx, s-1)$ preserves these vector fields. They therefore descend to M. Declare them orthonormal. Translation in s gives

$$D\psi^t E_u = e^{\lambda t} E_u \circ \psi^t, \quad D\psi^t E_0 = E_0 \circ \psi^t, \quad D\psi^t E_s = e^{-\lambda t} E_s \circ \psi^t. \tag{4}$$

This proves the singular-value formula and hyperbolicity.

For each positive integer m, B permutes $(m^{-1}\mathbb{Z}/\mathbb{Z})^2$. Rational points are therefore periodic and dense. Every suspended periodic orbit has expanding multiplier $e^{\lambda T} > 1$.

Finally, f is ergodic for Haar measure. Indeed, the Fourier coefficients of an invariant square-integrable function are constant along the orbits of B^T on $\mathbb{Z}^2$. Every nonzero orbit is infinite, so square summability makes every nonconstant coefficient zero. Ergodicity and full support provide a point whose forward orbit visits every member of a countable open basis infinitely often. Its suspended orbit has omega-limit set M, proving transitivity. □

3 The self-excited construction

Proof of Theorem 3. By Whitney's embedding theorem embed M smoothly in $\mathbb{R}^6$. Let E be its rank-three normal bundle, equipped with a metric connection. A sufficiently small disk bundle

$$E_R = \{(x, v) : x \in M,\ v \in E_x,\ \|v\| < R\}$$

is diffeomorphic to an open tubular neighborhood $U \subset \mathbb{R}^6$. Set $\kappa = 3\lambda/4$ and define a vector field on this bundle by

$$\dot{x} = E_0(x), \qquad \nabla_t v = -\kappa v. \tag{5}$$

Its flow is $(x, v) \mapsto (\psi^t x, e^{-\kappa t} P_x^t v)$, where P_x^t is metric parallel transport. The zero section A is a compact attracting invariant manifold, and the field has no equilibria.

At the zero section the derivative is the direct sum of $D\psi^t$ and $e^{-\kappa t} P_x^t$. For the orthogonal sum of g and the normal metric its ordered singular values are

$$e^{\lambda t}, \quad 1, \quad e^{-\kappa t}, \quad e^{-\kappa t}, \quad e^{-\kappa t}, \quad e^{-\lambda t}.$$

The adapted metric and the Euclidean metric are uniformly equivalent on the compact zero section. Consequently there is $C \geq 1$ such that the Euclidean exponents satisfy

$$|\nu_i(t, u) - \ell_i| \leq \frac{2 \log C}{t}, \qquad (\ell_i)_{i=1}^6 = (\lambda, 0, -\kappa, -\kappa, -\kappa, -\lambda), \tag{6}$$

uniformly in $u \in A$. This follows by comparing the norms at the source and target of the derivative; each singular value changes by at most a factor C^2.

The third and fourth partial sums of this spectrum are $\lambda/4$ and $-\lambda/2$. Uniform convergence, together with this strict crossing, gives

$$\mathrm{LD}(t, u) \longrightarrow 3 + \frac{\lambda/4}{3\lambda/4} = \frac{10}{3}$$

uniformly on A. It follows that $\dim_{\mathrm{L}} A = \mathrm{LD}(u) = 10/3$. The same crossing in $t^{-1} \log \omega_d$ proves $d_{\mathrm{L}}^E(u) = 10/3$. Also, for each j,

$$\lim_{t \to \infty} S_j(t, u) = \sum_{i=1}^{j} \ell_i = \lim_{t \to \infty} \max_{w \in A} S_j(t, w).$$

Taking $j = 3, 4$ proves (3).

It remains to make A self-excited without altering a neighborhood of A. Choose x_0 with omega-limit set M and $v_0 \neq 0$. The orbit through (x_0, v_0) approaches A and has omega-limit set A. Its normal radius decreases strictly. Choose a short flow box about this orbit, disjoint from A, with coordinates $(s, y) \in \mathbb{R} \times \mathbb{R}^5$, in which the field is $(1, 0)$. Choose the box using a section of constant normal radius. The original radius then depends only on s and decreases strictly, so the central orbit cannot return after its downstream exit.

Take $a > 0$ so that $|s| < 3a$, $\|y\| < 1$ lie in the box. Choose smooth cutoffs $0 \leq \eta, \zeta \leq 1$, with $\eta = 1$ for $|s| \leq 2a$, vanishing near $|s| = 3a$, and $\zeta = 1$ near $y = 0$, vanishing near $\|y\| = 1$. Write

$$\chi(s, y) = \eta(s)\zeta(y), \qquad H(s) = \frac{s^2 - a^2}{4a},$$

and replace the field by

$$\dot{s} = 1 + \chi(s, y)(H(s) - 1), \qquad \dot{y}_j = -j\chi(s, y)y_j \quad (1 \leq j \leq 5). \tag{7}$$

It agrees with $(1, 0)$ near the boundary and hence glues smoothly.

If $\chi = 0$, then $\dot{s} = 1$. At an equilibrium with $\chi > 0$, all transverse coordinates vanish. For $|s| \leq 2a$ the remaining equation is $H(s) = 0$, and for $|s| > 2a$ the velocity is a convex combination of two positive numbers. Thus the only equilibria are $p = (a, 0)$ and $q = (-a, 0)$, with Jacobians

$$DX(p) = \operatorname{diag}(1/2, -1, -2, -3, -4, -5), \qquad DX(q) = \operatorname{diag}(-1/2, -1, -2, -3, -4, -5).$$

The second is a sink; the first has dimension $1 + (1/2)/1 = 3/2$. At an equilibrium a coordinate change conjugates the Jacobian and does not change its asymptotic singular-value exponents.

The central line $y = 0$ is invariant. Its branch $s > a$ tends backward to p and exits the box in forward time. Afterwards it follows the original central orbit and approaches A with omega-limit set A. Hence every neighborhood of p meets $\mathcal{B}(A)$.

A smaller closed normal disk bundle misses the modification and remains a trapping neighborhood of A. All computations on A are unchanged. To verify forward completeness, choose $r_1 < R$ larger than every normal radius in the modification support. Above r_1, radius still decreases according to (5). Every trajectory therefore remains in a compact disk bundle of radius less than R, and the continuation theorem gives existence for all positive times.

Finally, transitivity implies inclusion-minimality. If a compact proper subset of A were a local attractor, a forward-dense orbit in A would enter its attracting neighborhood. Its omega-limit set would then lie in that proper subset, contradicting that the omega-limit set is A. □

4 An aperiodic invariant subset

Lemma 6. *There are a nonempty compact invariant subset $S_M \subset M$ with no periodic orbit and a smooth function $q : M \to [0, \lambda/4)$ such that $q^{-1}(0) = S_M$.*

Proof. The fixed point 0 of f has a nonzero transverse homoclinic point. To see this, decompose a nonzero integer vector as $m = u + s$, where u and s lie on the unstable and stable eigenlines. These lines have irrational slopes, so $u \bmod \mathbb{Z}^2 = -s \bmod \mathbb{Z}^2$ is nonzero. Its backward and forward iterates tend to 0, respectively. The intersection is transverse because the eigenlines are distinct.

The Smale–Birkhoff homoclinic theorem gives an integer $m_0 \geq 1$ and a compact set H on which f^{m_0} is conjugate to the full two-sided binary shift. Fix irrational $\alpha \in (0, 1)$ and take the closure Σ_α of the sequences

$$a_k(t) = \lfloor (k + 1)\alpha + t \rfloor - \lfloor k\alpha + t \rfloor, \qquad k \in \mathbb{Z}, \quad t \in \mathbb{R}.$$

This is a nonempty compact shift-invariant set. Telescoping shows that the number of ones in every block of length N differs from $N\alpha$ by at most one. The same holds in the closure. Thus every sequence has frequency α of the symbol one, and no sequence is periodic.

Let $K_0 \subset H$ be its image under the conjugacy and put $K = \bigcup_{j=0}^{m_0-1} f^j K_0$. Then K is compact and f-invariant. A periodic point in K would yield an f^{m_0}-periodic point in K_0, which is impossible. Its suspension S_M is compact, invariant and aperiodic: any suspension period must be an integer and would yield a periodic point of K.

To construct q, cover $M \setminus S_M$ by countably many relatively compact coordinate balls and choose nonnegative smooth bumps b_j whose positive sets cover that complement. Choose $c_j > 0$ with $\|c_j b_j\|_{C^j} \le 2^{-j}$. The series $F = \sum_j c_j b_j$ converges smoothly, vanishes precisely on S_M, and is nonnegative. Then

$$q = \frac{\lambda}{4} \frac{F}{1+F}$$

has all the required properties. □

5 The hidden construction

Proof of Theorem 4. Apply the smooth Nash embedding theorem to the metric in Lemma 5, obtaining an isometric embedding $\iota : (M, g) \hookrightarrow \mathbb{R}^N$. Work in $\mathbb{R}^{N+1}$, with tubular coordinates

$$\Theta(x, v, z) = (\iota(x) + v, z), \qquad v \in \nu_x \iota(M), \quad z \in \mathbb{R}.$$

Equip the normal bundle with its metric normal connection. Set $L = 2\lambda$ and $a = \lambda/2$, and define near the zero section

$$\dot{x} = E_0(x), \qquad \nabla_t v = -Lv, \qquad \dot{z} = -(a + q(x))z. \tag{8}$$

Writing $Q_t(x) = \int_0^t q(\psi^s x)\, ds$, the flow is

$$(x, v, z) \longmapsto \big(\psi^t x, e^{-Lt} P_x^t v, z e^{-at - Q_t(x)}\big). \tag{9}$$

Thus $A = \Theta(M, 0, 0)$ is a compact local attractor. Normal distance contracts at least as fast as e^{-at}. A sufficiently small closed tube is a trapping neighborhood whose maximal invariant set is A.

At the zero section the derivative of (9) is block diagonal. Derivatives of P_x^t and $Q_t(x)$ with respect to x are multiplied by v or z, and hence vanish there. The isometric embedding identifies the tangential and normal blocks with orthogonal Euclidean subspaces. Therefore the full Euclidean singular values are exactly

$$e^{\lambda t}, \quad 1, \quad e^{-at - Q_t(x)}, \quad e^{-\lambda t}, \quad \underbrace{e^{-Lt}, \dots, e^{-Lt}}_{N-3}. \tag{10}$$

They are in decreasing order, since $a \le a + Q_t(x)/t < 3\lambda/4 < \lambda < L$.

The first three exponent sums are positive at the crossing and the fourth is negative. Hence

$$\mathrm{LD}(t, \Theta(x, 0, 0)) = 3 + \frac{\lambda - a - Q_t(x)/t}{\lambda} = \frac{7}{2} - \frac{Q_t(x)}{\lambda t}. \tag{11}$$

At points of $S = \Theta(S_M, 0, 0)$, $Q_t = 0$ for every positive t. At any point outside S, positivity of $q(x)$ and continuity give $Q_t(x) > 0$. Thus

$$\max_{u \in A} \mathrm{LD}(t, u) = \frac{7}{2}, \qquad \{u \in A : \mathrm{LD}(t, u) = 7/2\} = S.$$

In particular $\dim_{\mathrm{L}} A = \max_{u\in A} \mathrm{LD}(u) = 7/2$.

Every periodic orbit $\gamma \subset A$ is disjoint from S, and compactness gives $q_\gamma = \min_\gamma q > 0$. Therefore

$$\mathrm{LD}(t,u) \le \frac{7}{2} - \frac{q_\gamma}{\lambda} < \frac{7}{2} \qquad (u \in \gamma,\ t > 0). \tag{12}$$

Its time average $\overline{q}_\gamma$ exists, so its local dimension equals $7/2 - \overline{q}_\gamma/\lambda$.

The exponential-growth definition gives the same local quantity at every point of A. Indeed, for $3 \le d \le 4$, (10) gives

$$t^{-1} \log \omega_d(D\varphi^t(\Theta(x,0,0))) = (7/2 - d)\lambda - Q_t(x)/t.$$

Its upper limit is $(7/2-d)\lambda - \liminf_t Q_t(x)/t$. The threshold lies strictly between 3 and 4, and hence

$$d_{\mathrm{L}}^E(\Theta(x,0,0)) = \frac{7}{2} - \lambda^{-1} \liminf_{t\to\infty} Q_t(x)/t = \mathrm{LD}(\Theta(x,0,0)).$$

No convergence of the average is needed away from periodic orbits and S.

It remains to arrange hiddenness and a complete global system. Put $\rho^2 = \|v\|^2 + z^2$ and choose $0 < r_0 < R$ within the tubular domain. Choose a smooth function h with $h = 0$ on $[0, r_0^2]$ and $h(R^2) = 3L$. In a slightly larger tube replace (8) by

$$\dot{x} = E_0(x), \qquad \nabla_t v = (-L + h(\rho^2))v, \qquad \dot{z} = (-a - q(x) + h(\rho^2))z. \tag{13}$$

There are no equilibria in this tube, since its base velocity is never zero. Along the boundary of $D = \Theta\{\rho \le R\}$,

$$\frac{d}{dt}\rho^2 = 4L\|v\|^2 + 2(3L - a - q(x))z^2 > 0.$$

Thus trajectories cannot enter D from its exterior.

Choose a smooth cutoff equal to one near D and supported in the slightly larger tubular domain. Use it to extend (13) to a smooth vector field on $\mathbb{R}^{N+1}$ equal to $-w$ outside a compact set. The resulting field has bounded derivative and at most linear growth, so is complete in both time directions. A large closed ball absorbs every bounded set. Its omega-limit set is a compact global attractor: forward images of the ball form nested nonempty compact sets, and their intersection attracts the ball uniformly and therefore attracts every bounded set.

All equilibria are outside D. Every point approaching $A \subset \operatorname{int} D$ must eventually enter D, whereas no exterior trajectory can do so. Hence $\mathcal{B}(A) \subset \operatorname{int} D$. Each equilibrium has an exterior neighborhood disjoint from that basin, proving hiddenness. If desired, translate the embedding so that the cutoff support avoids the origin; then the origin is a sink of the extended field.

The extension leaves a neighborhood of A unchanged. Lemma 5 gives transitivity and dense unstable periodic orbits, and the stable normal directions preserve hyperbolicity. Inclusion-minimality follows by the same dense-orbit argument as in Theorem 3. Finally, (12) excludes every periodic orbit from the maximizing set. □

6 Consequences

Theorem 3 gives a self-excited attractor satisfying the common-critical-point condition for which no unstable equilibrium supplies the proposed dimension bound. The conclusion remains false if the eligible equilibrium is additionally required to have an unstable branch approaching and tracing the attractor.

Theorem 4 gives a hidden attractor with aperiodic local-dimension maximizers despite dense unstable periodic orbits. The gap may depend on the periodic orbit; no uniform positive gap over all periodic orbits is asserted. These are local-attractor statements. Neither substitutes the local attractor for the global attractor in the historical stationary-or-periodic maximization assertion.

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References

[1] N. V. Kuznetsov and V. Reitmann, *Attractor Dimension Estimates for Dynamical Systems: Theory and Computation*, Springer, 2021.

[2] N. V. Kuznetsov, *The Lyapunov dimension and its estimation via the Leonov method*, arXiv:1602.05410.

[3] V. Araújo and V. Pinheiro, *Abundance of wild historic behavior*, arXiv:1609.05356, Section 4.7.1.

[4] J. Nash, The imbedding problem for Riemannian manifolds, *Annals of Mathematics* **63** (1956), 20–63.

[5] H. Whitney, The self-intersections of a smooth n-manifold in $2n$-space, *Annals of Mathematics* **45** (1944), 220–246.